

Project Report: Color Vision Correction System

1. Project Overview

Project Name: Bring your own project (AI Color Correction)

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Course: CSE(AI ML)

Objective: To bridge the gap between digital design and accessibility using Deep Learning.

2. Problem Statement

Color Vision Deficiency (CVD) affects a significant portion of the population. Traditional methods of correction (like Dalmoralization) use fixed mathematical matrices which can sometimes lead to "washed out" images. This project aims to use a more sophisticated **U-Net** approach to retain image quality while maximizing contrast.

3. Tools & Technologies Used

Tool	Purpose
Python	Core programming language.
OpenCV (cv2)	Image preprocessing and real-time video stream handling.
U-Net (TensorFlow/PyTorch)	Deep learning architecture for image-to-image color translation.
NumPy	High-speed matrix operations for pixel manipulation.
Gradio	Deploying the model into a functional, browser-based GUI.
Matplotlib	Visualizing color histograms and training loss curves.

4. Technical Workflow

1. **Dataset Preparation:** Utilizing datasets of original images paired with simulated versions of Protan, Deutan, and Tritan views.
2. **Model Training:** Training the **U-Net** to recognize the "lost" data in simulated images and learn the transformation back to a high-contrast state.
3. **Optimization:** Using `opencv-python-headless` for efficient backend processing without unnecessary GUI overhead.
4. **Deployment:** Using Gradio to create sliders for the user to toggle between different types of color blindness and correction intensity.

5. Future Scope

- Integrating the model into a Chrome Extension for real-time web browsing.
- Developing a mobile-friendly version for use with smartphone cameras.